Science

Nature Walks & Scouting
General Science
Labs
Natural History
Nature Notebook

Alveary

Science: Grade 7

GRADE(S): 7

About the Course

In level 7, learners extend their relationship with science in a new direction (time), situating science in its historical, political, and cultural context, and begin to understand that science is an ever-changing process rather than a static body of knowledge. Reading level and expectations increase from previous levels.

This course includes the following topic(s): General Science: Grade 7, Natural History: Grade 7, Labs: Grade 7, Nature Notebook: Grade 7, Nature Walks & Scouting: Grades 1-8

About Nature Walks & Scouting: Grades 1-8
Students spend time once each week on a longer excursion. Prompts can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

About General Science: Grade 7
This is the first of a two-year Physical Science course (i.e., introductory physics and chemistry). The fullness of the course includes topics in earth science, history, politics, and more. The completion of this course provides a gentle yet robust transition into the high school disciplines. The lab component is required for this course. Coordinating afternoon activities are provided in Outdoor Work.

About Labs: Grade 7
This is the required lab component for level 7 General Science. This course includes our custom laboratory book which will build foundational and progressive skills and habits.

About Natural History: Grade 7
Learners dig deeper into relationships between creatures as they read about the beginnings of ecology. Coordinating afternoon activities are provided in Outdoor Work.

About Nature Notebook: Grade 7
Students spend time at least once each week recording their observations. Prompts that coordinate with curricular content can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

Science: Grade 7
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. In Form 1, students became acquainted with this area of knowledge, science. Form 2 students extended their scope and began to encounter science as an active part of society. Form 3 students extend their relationship with science in a new direction (time) and consider science as a process by which man seeks Truth within Creation. They begin to situate science in its historical, political, and cultural context with all of the complexities of man's own story. In this way, students are prepared to think about science with a complete and holistic perspective.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

Science: Grade 7
Learners may stay at their appropriate level for General Science while combining Natural History and Outdoor Work, or teachers can adapt the laboratory content and reading level to meet their needs.

Nature Walks & Scouting: Grades 1-8
Learners may be combined and share prompts or follow the plan of their local scouting troop or natural history club. Support accessibility and sensory development as appropriate.

General Science: Grade 7
For seventh-grade students or possibly hungry sixth-graders or eighth-graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, or teachers can adapt the reading level and laboratory content to meet their needs.

Labs: Grade 7
The ideas and skills in this component are progressive, like math or grammar. Teachers should read the lab book thoroughly to understand what concepts might need to be supplemented should they choose a different sequence or a substitution.

Natural History: Grade 7
For approximately seventh-grade students, based on a balance of complexity and reading level. However, learners may be freely combined based on needs and preferences.

Nature Notebook: Grade 7
Learners may be combined and share prompts or follow their own interests. Support through varied media, scribing observations, or notebooking in tandem.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
1-8	Nature Walks & Scouting: Grades 1-8 1 time/week 30 min+	
7	General Science: Grade 7 2 times/week 30 min	The Story of Science: Aristotle Leads the Way
7	Labs: Grade 7 1 time/week 45 min	Science: Grade 7 Lab Book
7	Natural History: Grade 7 1 time/week 30 min	The Girl Who Drew Butterflies: How Maria Merian's Art Changed Science
7	Nature Notebook: Grade 7 1+ time/week 20 min+	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 7				
Natural History: Grade 7	General Science: Grade 7	General Science: Grade 7	Nature Notebook: Grade 7	Labs: Grade 7 Nature Walks & Scouting: Grades 1-8

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 7
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings, except where the suggestion is to collect some Thing for a lesson.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 7
Encourage students to choose their own special topic and to notice its ecological relationships. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too!

☐ Learn a few of your local species or varieties in connection with special topics.

General Science: Grade 7
☐ Terms 1 and 3: a planetarium or local astronomy club event
☐ Term 2: a playground

Natural History: Grade 7
☐ Any term: natural history museum or butterfly house

Term Prep & Teacher Tips

General Science: Grade 7
☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
- medium-large sized recycled box, roughly 1-2'
- knife, blade, or scissors to cut the box
- table/reading lamp
- table or countertop
- books or wood blocks to adjust height
- 1 gallon jug of milk or water, the type with a handle
- horizontal bar to support weight, such as broom handle held by 2 people. a chin up bar, a porch railing, etc.
- recycled cereal box or toothbrush
- a flat location that is in the sun at noon
- various supplies by student design
- printables
- computer
- optional: empty cereal box
- optional: orange
- optional: flashlight
- optional: calculator, such as found on most electronic devices

Natural History: Grade 7
☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
- a flower with large, visible parts, such as a lily (Term 1)
- a sunny window
- water
- a few sprigs of any water weed obtained from a pond or pet store that sells fish (Term 3)
- various supplies by student design
- optional: salt
- optional: forceps/tweezers

Reminders

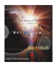
Natural History: Grade 7
☐ Note that the first lesson of Natural History is an activity, so be prepared with these and any other materials from the supply list on day 1.

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.
∞ [View Book List Details]

Science: Grade 7

General Science: Grade 7


The Story of Science: Aristotle Leads the Way

Labs: Grade 7

Science: Grade 7 Lab Book

Natural History: Grade 7


The Girl Who Drew Butterflies: How Maria Merian's Art Changed Science

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.
∞ [View Basic Supplies]
∞ [View Supply List Details]

Science: Grade 7

Alveary Grade 7 Science Kit

Nature Walks & Scouting: Grades 1-8


Wild Bird Seed


Bird Feeder


Hand Lens


Small Collection Containers


Small Basin


Dip Net


Student Microscope

General Science: Grade 7

Household Items - General Science: Grade 7

Natural History: Grade 7

Household Items - Natural History: Grade 7

 Slides and Coverslips

 Prepared Microscope Slides

 Student Microscope

Quick Links

Click THIS text or scan the QR code for links.

Science: Grade 7

- Alternate One Year Physical Science Lesson Plans
- Outdoor Work
- Seek app from iNaturalist
- SkyView Lite for Android
- SkyView Lite for iOS

Labs: Grade 7

- Grade 7 Lab Book
- Lab Notebook Examples

Science: Grade 7

How To Teach

Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there were any dates that they want to keep for their Book of Centuries.

Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

- These can be viewed as alternative ways to engage.

Science: Grade 7

How To Teach

Introduce

Regardless of how many days are required to complete a particular activity, every science lab has the same flow, which follows the scientific method and is guided by the lab book.

• On day 1, learners read an introduction in their lab book. How does this lab activity relate to what they have learned so far, as well as any previous experience? What will they learn from the lab? This is analogous to the conversation we might have when we begin something new in any subject, but they may need to dialogue as they learn to extend these skills to the laboratory.

• Once they have had a chance to think, they will compose a prelab narration to put these introductory ideas into their notebooks. The prompt in the lab is generalized and consistent, so they learn the habit. Eventually, they will learn to formulate this as a hypothesis. For example, a learner preparing for a lab about the use of insect repellent might write:

"I have read about some diseases that are spread by insects, like Lyme disease. I also know that my sister is allergic to some insect bites. Insect repellent contains pesticides to keep insects away. Some scientists worry about how pesticides affect wildlife. I am going to compare some different insect repellants in this lab to see if they really work."

• Written narration and composition are skills that they will build over time. These prelab narrations may seem short and even incomplete at first, but that is okay. If learners have difficulty or are easily frustrated, then provide them with support. Teachers may act as scribes or allow students to keep a digital notebook to type or use assistive technology, as appropriate.

• After they complete their prelab narration, the listed materials are collected. This gives the learner some responsibility to let the teacher know if something is missing or to remind the teacher if something needs to be purchased at the store.

• If these activities on day 1 do not fill the scheduled time, that is fine. They might use the additional time to familiarize themselves with the procedure, draw a picture from their book, or catch up on any other work. Some labs may instruct the student to begin on the same day.

Lab Procedure

• Then, students begin and follow the procedure (whenever prompted by the lab book). Note that the lesson plans guide teachers as to which sections are completed each week, and the lab book instructs learners when to take a break.

• The lab gives instructions for using their notebooks to create tables and figures, as needed. Do not allow this to become an obstacle. Do it with them, first having them watch and then having them copy or help when ready.

• If teachers choose to have learners record directly in the lab book or on a photocopy, then cut out and tape these into their lab books, so that they can see how the record is built.

• Learners may feel unsatisfied with their results. This is often part of the process. Come alongside, help them learn to be okay with uncertainty and questions, and teach them what to do with it in the next step.

Analysis & Conclusions

• The last step in the lab is to analyze the data and observations and draw conclusions from them. For some simpler labs, learners will complete their analysis and concluding (or postlab) narration on the last day of the lab procedure. For labs that are more involved, a separate day has been built in to allow adequate time for this.

• Similar to the prelab narration, the concluding narration is a chance to think about and put into words (or questions). This time, they are considering what they learned from the lab, what they could learn more about if they were to continue, and possibly how they would pursue that learning. Again, they might need support in the form of dialogue, a scribe, etc. For example, the above learner might write:

"It was clear that the Off and black pepper essential oil worked against ants because they would not even touch the line of repellent, but I wasn't sure about the Skin So Soft because it ran all over the place. I could test this one again with different insects or on a different surface."

• Depending on the interest of the learner and the priorities of the teacher, the student might be encouraged to spend more time on those ideas of what more they could learn, or it might be time to move on. Either way, it is an important part of the scientific method to reflect on what we could or would do next - our practice should help to clarify our thinking and teach that there is always more to be learned.

• Teachers should engage learners with this reflection by reviewing their lab notebooks with them, discussing the science used in the lab, and demonstrating curiosity about the lab themselves.

Science: Grade 7

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 1

30m Natural History: Grade 7 - Lesson 1

Shoots Activity

> Materials: 9 bean seeds, 3 recycled pots or cups, soil of any kind, aluminum foil (for later), sunny window, recycled box to hold 3 pots or cups, water

→ INTRO
This term, you are going to read the biography of a woman who began as a young artist and grew to be a skilled naturalist and biologist. We will begin as she did, studying the anatomy and physiology of flowering plants. Today, you will plant some bean seeds and observe how they "see" the light.

→ ACTIVITY
• Prepare 3 pots or cups with soil and plant 3 seeds in each cup.
• Set the cups near a sunny window, but inside a recycled box on its side, such that they only receive light from the direction of the window. Keep them watered so that the soil is damp, but not wet. They may only need 1-2 tablespoons of water at a time, depending on the size of your cups. It will take them 1-2 weeks to germinate, and you will need to keep the soil damp.

→ NARRATE
Draw or diagram your planted seed cups in their window in your science notebook.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1

30m General Science: Grade 7 - Lesson 1

Thinking About Creation

> Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ INTRO
This book tells the story of how modern science developed, beginning with the ideas of the ancient Greeks. Most of them wondered and learned from the heavens and from mathematics, eventually defining science. There are many interesting sidebars and tangents in the text, but read the main text first and then explore them, as desired. Watch the course introduction video:
∞ Video Link: Form 3 Science Intro (5:04)

→ READ, NARRATE, & DISCUSS
Ch.1 p.1-8 "The universe" - "to have them."
• Tell about the earliest beginnings of science in the ancient world. How did men begin?
• The author of this book invites all learners of different beliefs to find common ground in this discussion about how we understand the world we share. Discuss some of the ways she does this.

→ SUPPLEMENTAL
∞ Video Link: NatGeo Mesopotamia (4:10)
∞ Video Link: Mayan Creation Story (2:55)
∞ Video Link: Hindu Creation Story (2:41)

★ TEACHER TIP
Watch the course introduction video with your student(s).

★ TEACHER TIP
Even as students begin to work more independently, teacher engagement is very important. Discussion prompts are provided to help, if natural discussion is a challenge.

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• IMPORTANT DATES
Sumerian Empire (c.3100 B.C.-c.2000 B.C.), Egyptian Civilization (c.3100 B.C.-332 B.C.)

Science: Grade 7

Click THIS text or scan the QR code for links.

Term 1

WEEK 1

30m General Science: Grade 7 - Lesson 2
What is a Myth?

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.2 p.9-14 "Once, or so" - "makes us human."
- According to the author, what is the distinction between myth and science? And what is NOT the distinction? Is there anything to add based on your own tradition?
- Why is myth important? If you have read much of J.R.R.Tolkien's work, consider the role myth plays in his stories. Compare/contrast both authors' ideas about myth. If you haven't yet read his work, consider another author who makes use of myth, or file away this idea for the day that you do read his work!
- What is a hypothesis?
- How is science (and its limits) part of what makes us human?

→ SUPPLEMENTAL
∞ Video Link: Finding Summer Constellations (5:39)

★ TEACHER NOTE
The author invites readers to consider what is myth and what is science, a consideration that engages with religion, on p.11-12: "How do we" - "fiction usually does." Teachers might choose to allow the idea to pass by the way or to engage in discussion.

WEEK 1

45m Labs: Grade 7 - Lesson 1
Changing Constellations

Materials: Grade 7 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 1 of Changing Constellations, as directed in the Lab Book.

Suggested day 1: Students read the Introduction. Then compose the prelab narration in the lab notebook. These need not be more than 1-3 sentences at first, and teachers should feel free to scribe for students, as necessary. Spend the remaining time gathering materials for next week. If time permits, students may want to copy any helpful diagrams into their notebooks.

★ TEACHER TIP
Students study sky charts over different time periods to notice how observed movement of stars changes. Remember the most important objective is always building skills and habits. Pacing is only a suggestion. Students should engage with lab at a pace that is appropriate for their abilities and interest.

WEEK 2

30m Natural History: Grade 7 - Lesson 2
The Girl in the Garden

Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

→ INTRO
Let's begin reading about Maria Merian today. As we continue to learn about her and her work, we will see the interactions between plants, insects, and the environment. These shared relationships exist for all living things around the world and are called the study of ecology.

→ READ, NARRATE, & DISCUSS
"Prologue" p.viii-ix "A girl kneels" - "her story.

★ TEACHER TIP
The next 2 weeks are activity weeks!

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Science: Grade 7

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 2 ⬚ **30m General Science: Grade 7 - Lesson 3**
What is a Myth?

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.2 p.14-19 "We've learned" - "to be born."
• Describe the process of science.
• "The world is God's epistle written to mankind… It was written in mathematical letters." Discuss.
• The author says that the "marriage of numbers and nature… is an essential part of the scientific story." Why? Do you agree? Where have you noticed numbers in nature?
• Why do you suppose the story begins with sky watching?

→ SUPPLEMENTAL
∞ Video Link: Is Math Invented or Discovered? (5:11)

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 2 ⬚ **30m General Science: Grade 7 - Lesson 4**
Thinking about Time

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.3 p.20-25 "Perhaps it was" - "around the globe."
• What kinds of observations and what kinds of questions did early sky watchers have?
• Explain the similarities and differences between the lunar calendar and the solar calendar. Which is closer to the calendar we use?

→ SUPPLEMENTAL
∞ Video Link: The Analemma (4:23)
∞ Video Link: Want to measure the analemma? (7:33)

• IMPORTANT DATES
Egyptian Solar Calendar (c.4th century B.C.)

WEEK 2 ⬚ **45m Labs: Grade 7 - Lesson 2**
Changing Constellations

Materials: Grade 7 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of Changing Constellations, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today, using the website in the lab book.

WEEK 3 ⬚ **30m Natural History: Grade 7 - Lesson 3**
Shoots Activity

Materials: aluminum foil, water

→ INTRO
You should have shoots by now, and they are probably growing rapidly! Today, you will change their conditions to see if they respond.

→ ACTIVITY
• Cut 3 small squares of aluminum foil (about 2" x 2") and 3 small

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Term 1

rectangles (about 1/2" x 3").
• Form the squares over your fingertip or a pencil to form 3 small caps.
• Wrap the rectangles around your finger or pencil to form 1/2" tall tubes.
• Place the caps on the tips of the seedlings in one cup; place the tubes around the bases of the seedlings in another cup. Leave the third cup alone - this is your control. Continue watering your cups this week.

→ NARRATE
Draw or diagram your seedlings in your science notebook.

WEEK 3

30m General Science: Grade 7 - Lesson 5
Phases of the Moon

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.3 p.26-27 "For a long time," - "facing the sun."
• Explain the moon's phases.

→ SUPPLEMENTAL
∞ Video Link: Moon Phases (9:45)

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 3

30m General Science: Grade 7 - Lesson 6
Thinking about Time

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.3 p.28-31 "Palenque," - "our solar system."
• Were the ancient calendar makers doing science? Why or why not? If not science, what was it?
• What ideas have been passed down to us from the ancient calendar makers?

→ SUPPLEMENTAL
∞ Video Link: Decoding the Astronomy of Stonehenge (6:29)

• FIELD TRIP
Visit a planetarium or astronomy club event.

WEEK 3

45m Labs: Grade 7 - Lesson 3
Changing Constellations

Materials: Grade 7 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 3 of Changing Constellations, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 3: Students complete the Analysis and Conclusion.

★ TEACHER TIP
Inviting students to share their lab notebooks is a great way to engage in discussion and keep up with their learning! They should be able to discuss the science and explain what they did.

Science: Grade 7

Click THIS text or scan the QR code for links.

Term 1

WEEK 4 ⬚ **30m Natural History: Grade 7 - Lesson 4**
Shoots Activity

Materials: plantings from earlier in the term

→ INTRO
See how your shoots responded to their foil covers.

→ OBSERVE
Carefully observe how each of your seedlings grew this week. Are there differences? Does this reveal anything about how plants "see" light?

→ NARRATE
Draw or diagram your seedlings in your science notebook.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 4 ⬚ **30m General Science: Grade 7 - Lesson 7**
Ancient Record Keeping

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.3 p.32-33 "When did people" - "closely guarded secret."
• Discuss the beginning of math.

→ SUPPLEMENTAL
∞ Video Link: Ancient Writing and Modern Tech (6:44)

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 4 ⬚ **30m General Science: Grade 7 - Lesson 8**
Thinking about Nature

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.4 p.34-38 "Does where we live" - "scientific approach."
• Set the scene for science in Ionia.
• How was the observation of patterns in nature an important concept?

• IMPORTANT DATES
Thales (c.636 B.C.-546 B.C.)

WEEK 4 ⬚ **45m Labs: Grade 7 - Lesson 4**
A Changing Moon Activity

Materials: Grade 7 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 1 of A Changing Moon Activity, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction and begin the Procedure this week.

★ TEACHER TIP
Students create a model to understand the phases of the moon.

Science: Grade 7

Click THIS text or scan the QR code for links.

Term 1

WEEK 5

30m Natural History: Grade 7 - Lesson 5

Egg

Materials: The Girl Who Drew Butterflies

→ RECAP
What did you read or observe last week?

→ READ, NARRATE, & DISCUSS
Ch.1 p.1-6 "Maria Sibylla" - "imagination instead?"
• Why do you think the author titled the first chapter "Egg"? Notice the chapter titles throughout the book.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5

30m General Science: Grade 7 - Lesson 9

Thinking about Nature

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.4 p.38-41 "We don't know" - "is heating up."
• Why was Thales important?
• A recurring idea: how is an incorrect idea still important?
• "While others trembled at nature's wonders and puzzles, the Ionians asked hard questions." Discuss.

→ SUPPLEMENTAL
∞ Video Link: Law vs. Theory (5:12)

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 5

30m General Science: Grade 7 - Lesson 10

Solving Real-Life Problems with Math

Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.4 p.42-43 "The eye can" - "about 350 paces."
• Explain how one can measure from a distance.

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in the habits of observation, measurement, and/or record-keeping? More interest/ease in discussing the challenges of early scientists in their social context? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a planetarium. Reach out through Contact Us, if you need help.

WEEK 5

45m Labs: Grade 7 - Lesson 5

A Changing Moon Activity

Materials: Grade 7 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 2 of A Changing Moon Activity, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 2: Students complete the Procedure, as well as Analysis and Conclusions, this week.

★ TEACHER TIP
Be sure to have students share their notebooks, during which they should be able to discuss the science and explain what they did.

Science: Grade 7

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 6

30m Natural History: Grade 7 - Lesson 6
Hatching

Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

→ RECAP
What did you read or observe last week?

→ READ, NARRATE, & DISCUSS
Ch.2 p.7-12 "Before Maria" - "about them."

★ TEACHER TIP
Are your learners demonstrating more familiarity/interest in plants? More interest/ease in discussing the relationships that plants have with animals and the rest of the environment? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a nature preserve or natural history museum.

● OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6

30m General Science: Grade 7 - Lesson 11
Dangerous Ideas

Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.5 p.44-49 "Thales did" - "started so well."
- The author states that scientific questioning "was threatening to the keepers of the old ideas." Why do people sometimes feel threatened by new ways of thinking?
- What kinds of ideas did the scientists in this chapter think about?
- What was the danger of Anaxagoras' ideas? Can science be dangerous? If so, under what circumstances?

★ TEACHER TIP
Remember to engage students on their reading through natural discussion. If you need help, use the bulleted prompts.

★ TEACHER NOTE
Evolution is mentioned in the third paragraph on p.44. Teachers might choose to allow it to pass by the way or to discuss with students.

● IMPORTANT DATES
Anaximander (c.611 B.C.-c.547 B.C.), Anaximenes (c.570-500 B.C.), Anaxagoras (c.500-428 B.C.)

WEEK 6

30m General Science: Grade 7 - Lesson 12
The Origins of Math

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.5 p.50-53 "It was the" - "in Egyptian tombs."
- Discuss how the concept of zero changed mathematics.

WEEK 6

45m Labs: Grade 7 - Lesson 6
A Curved Earth

Materials: Grade 7 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 1 of A Curved Earth, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction and begin the Procedure this week.

★ TEACHER TIP
Students use sky charts to support or refute the hypothesis that the Earth is curved.

Science: Grade 7

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 7

⬚ **30m Natural History: Grade 7 - Lesson 7**
First Instar

▯ Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

→ RECAP
What did you read or observe last week?

→ READ & NARRATE
Ch.3 p.13-17 "Art was a joy" - "for profit."

→ READ, NARRATE, & DISCUSS
p.18-19 "Art was big" - "busy workshop."

→ STUDY
Maria had to study every detail of the plants she learned to paint. Study the diagram at the link and try to learn the parts of the leaf. Copy the diagram, if helpful.
∞ Image Link: Leaf Diagram

★ TEACHER TIP
Some students will benefit from reading and narrating the selection in parts because the two passages have different purposes. Can they identify the purpose of each one?

● OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7

⬚ **30m General Science: Grade 7 - Lesson 13**
Elements

▯ Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.6 p.54-57 "Thales said" - "a small country."
● What's the 'big idea' or 'main purpose' of this chapter? Use the title to help, if it's not immediately clear.
● How does this chapter connect back to ideas from previous chapters? At the same time, how does it move the story forward?

★ TEACHER TIP
Help students begin to discern the main ideas of a chapter and link this to previous information.

● NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

● IMPORTANT DATES
Empedocles (c.495 B.C.-c.434 B.C.)

WEEK 7

⬚ **30m General Science: Grade 7 - Lesson 14**
Thinking at Sea

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.7 p.58-61 "Herodotus" - "of tall tales."
● Tell about Herodotus.
● How did sea exploration add to scientific observation?
● Discuss how the sailors and Herodotus were both engaging in scientific thinking.

WEEK 7

⬚ **45m Labs: Grade 7 - Lesson 7**
A Curved Earth

▯ Materials: Grade 7 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

★ TEACHER TIP
Be sure to have students share their notebooks, during which they should be able to discuss the science and explain what they did.

Science: Grade 7

[Click THIS text or scan the QR code for links.]

Term 1

Complete day 2 of A Curved Earth, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 2: Students complete the Procedure, as well as Analysis and Conclusions, this week.

WEEK 8

⬚ **30m Natural History: Grade 7 - Lesson 8**

Second Instar

▯ Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

→ RECAP
What did you read or observe last week?

→ READ, NARRATE, & DISCUSS
Ch.4 p.20-22, 24-27 "Maria loved" - "her work."

→ READ, NARRATE, & DISCUSS
p.23 "Today" - "to record it."

★ TEACHER TIP
Note that students will need a flower with large, visible parts next week. A variety of lilies works well.

● **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8

⬚ **30m General Science: Grade 7 - Lesson 15**

Polestars

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.7 p.62-63 "The North Star" - "show the way."
● Tell what you know about pole stars.

★ TEACHER NOTE
The timescale of Earth's existence is mentioned on p.63: "takes about 25,800 years." Teachers might choose to allow it to pass by the way or to discuss alternative theories.

WEEK 8

⬚ **30m General Science: Grade 7 - Lesson 16**

Science and Math

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.8 p.64-69 "A Sense of Number" - "change the world."
● Tell the story of the birth of Pythagoras.
● Discuss the importance of math.
● Try solving magic squares. The magic numbers for some squares are as follows: 4×4 is 34, 6×6 is 111, 7×7 is 175, 9×9 is 369. Can you find the number for a 5×5 square?

● **IMPORTANT DATES**
Pythagoras (c.582 B.C.-507 B.C.)

WEEK 8

⬚ **45m Labs: Grade 7 - Lesson 8**

Moon and Tides

▯ Materials: Grade 7 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 1 of Moon and Tides, as directed in the Lab Book.

★ TEACHER TIP
Students use tide data to evaluate any connection to lunar phases.

Science: Grade 7

[Click THIS text or scan the QR code for links.](#)

Term 1

Suggested day 1: Students complete the Introduction. Spend the remaining time gathering materials for next week. If time permits, students may want to copy any helpful diagrams into their notebooks.

WEEK 9

⬚ **30m Natural History: Grade 7 - Lesson 9**

Flower Dissection Activity

> ⎕ Materials: image, video, and article links, a flower with large, visible parts, optional: tweezers or toothpick

PREP: Read Teacher Tip

→ INTRO
Today, you will dissect a flower to examine all the parts carefully.

→ OBSERVE
Begin from the outside of the flower and work your way inward, carefully observing and removing each part. You can refer to the "Flower Anatomy" diagram at the link if desired.
∞ Image Link: Flower Anatomy

→ NARRATE
Draw or diagram your flower in your science notebook.
•Try to explain pollination using your diagram. On which part is the pollen? Where do the seeds or fruit develop?

→ SUPPLEMENTAL
You can learn how the seeds are fertilized and develop into fruit at the following links:
∞ Video Link: Germination
∞ Article Link: Seed and Fruit Development

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9

⬚ **30m General Science: Grade 7 - Lesson 17**

Pi

> ⎕ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.8 p.70-71 "Of all the" - "the ancient world."
• What is pi and why is it elusive?
• In the video, note what a challenge it can be to find the source of bias in an experiment and eliminate it! Try playing with pi. Knowing that circumference equals pi times diameter, find various objects for which you can measure either circumference or diameter, and then calculate the other. If the circumference of the Earth is 24,901 miles, what is the diameter?

→ SUPPLEMENTAL
∞ Video Link: Finding Pi with Darts (5:48)

WEEK 9

⬚ **30m General Science: Grade 7 - Lesson 18**

Pythagoras

> ⎕ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.9 p.72-75 "Samos was part" - "if it really happened."

Science: Grade 7

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Term 1

• What kinds of data have these early scientists been collecting so far? What do the Greeks bring? Is anything still wanting?
• Pythagoras "thought numbers were... an expression of God's mind." Discuss. Is mathematics a language? If so, does God use this language?

WEEK 9 ⬚ **45m Labs: Grade 7 - Lesson 9**
Moon and Tides

▯ Materials: Grade 7 Lab Book and materials listed within

➔ LAB DAY
Complete day 2 of Moon and Tides, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today, using the website and constructing tables.

WEEK 10 ⬚ **30m Natural History: Grade 7 - Lesson 10**
Third Instar

▯ Materials: The Girl Who Drew Butterflies

➔ RECAP
What did you read or observe last week?

➔ READ, NARRATE, & DISCUSS
Ch.5 p.28-33 "Maria continued" - "was to marry."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10 ⬚ **30m General Science: Grade 7 - Lesson 19**
Pythagoras

▯ Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.9 p.75-81 "Remember, the" - "has done more."
• Were Pythagoras' ideas good? Were they dangerous? Why or why not? Good or dangerous, were they important?
• How does this chapter connect back to ideas from previous chapters? At the same time, how does it move the story forward?
• Notice the shapes toward the bottom of the article and try drawing your own fractals from any starting line, curve, or shape!

➔ SUPPLEMENTAL
∞ Video Link: How to Prove the Pythagorean Theorem (5:17)
∞ Article Link: Fractals in Nature

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 10 ⬚ **30m General Science: Grade 7 - Lesson 20**
Thinking about Irrational Numbers

▯ Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.9 p.82-85 "The Pythagoreans" - "in a pentagram."
• Draw a picture that shows or explains the importance of an irrational number.

Science: Grade 7

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Term 1

➔ SUPPLEMENTAL
∞ Video Link: Making Sense of Irrational Numbers (4:41)

WEEK 10 ⬚ **45m Labs: Grade 7 - Lesson 10**
Moon and Tides

Materials: Grade 7 Lab Book and materials listed within

PREP: Read Teacher Tip

➔ LAB DAY
Complete day 3 of Moon and Tides, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 3: Students complete the Analysis and Conclusion.

★ TEACHER TIP
Be sure to have students share their notebooks, during which they should be able to discuss the science and explain what they did.

WEEK 11 ⬚ **30m Natural History: Grade 7 - Lesson 11**
Catch Up

Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

➔ CATCH UP
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

★ TEACHER TIP
When students take exams next week, evaluate what they recall, but more importantly, whether plants or plant relationships in the ecosystem are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

● OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11 ⬚ **30m General Science: Grade 7 - Lesson 21**
Thinking about Atoms

Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.10 p.86-91 "I would rather" - "believed in atoms."
● Discuss how Democritus took the idea of elements further.
● Explain the central idea of Democritus' atom.
● Discuss how science moves beyond an initial idea and explain why the ancient Greeks could not move forward.

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly whether the practical habits of science or the challenges of early scientists are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? Reach out through Contact Us, if you need help.

● IMPORTANT DATES
Democritus (c.460 B.C.-c.370 B.C.), Socrates (469-399 B.C.), Aristotle (384-322 B.C.)

WEEK 11 ⬚ **30m General Science: Grade 7 - Lesson 22**
A Poem on the Nature of Things

Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

Science: Grade 7

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Term 1

	➔ READ, NARRATE, & DISCUSS Ch.10 p.92-93 "Lucretius" - "one oxygen atom." • What kinds of observations does Lucretius note in his poem?	
WEEK 11	⬚ **45m Labs: Grade 7 - Lesson 11** *Catch Up* Materials: Grade 7 Lab Book and materials listed within **➔ CATCH UP** Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	
WEEK 12	⬚ **30m Natural History: Grade 7 - Lesson 12** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: The Girl Who Drew Butterflies	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 12	⬚ **30m General Science: Grade 7 - Lesson 23** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: Aristotle Leads the Way	
WEEK 12	⬚ **30m General Science: Grade 7 - Lesson 24** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: Aristotle Leads the Way	
WEEK 12	⬚ **45m Labs: Grade 7 - Lesson 12** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: Grade 7 Lab Book	

Science: Grade 7

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Term 2

WEEK 1 ⬚ **30m Natural History: Grade 7 - Lesson 13**
Fourth Instar

📖 Materials: The Girl Who Drew Butterflies

→ RECAP
Recall where we left Maria Merian last term.

→ READ & NARRATE
Ch.6 p.34-38, 40 "At the age" - "for both."

→ READ, NARRATE, & DISCUSS
p.39 "Nearly 100,000" - "in society."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1 ⬚ **30m General Science: Grade 7 - Lesson 25**
Aristotle

📖 Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ RECAP
Where did the story leave off? Look back if you need help.

→ READ, NARRATE, & DISCUSS
Ch.11 p.94-97 "By the time" - "those ideas well."
• What's the 'big idea' or 'main purpose' of this chapter? Use the title to help, if it's not immediately clear.
• Explain Plato's philosophy.

★ TEACHER TIP
Remember to engage students on their reading through natural discussion. If you need help, use the bulleted prompts.

• IMPORTANT DATES
Plato (c.427 B.C.-347 B.C.)

WEEK 1 ⬚ **30m General Science: Grade 7 - Lesson 26**
Plato's Search for Perfection

📖 Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.11 p.98-99 "Plato looked" - "in understanding them."
• Try using cardboard or cardstock to make the 5 Platonic solids. How many of Archimedes' solids can you render or imagine?

WEEK 1 ⬚ **45m Labs: Grade 7 - Lesson 13**
Engineering a Catapult

📖 Materials: Grade 7 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 1 of Engineering a Catapult, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction. Spend the remaining time gathering materials for next week. If time permits, students may begin steps 1-2 of the Procedure or may wait until next week.

★ TEACHER TIP
Students construct and test a catapult to learn about a simple machine. Remember, the most important objective is always building skills and habits. Pacing is only a suggestion. Students should engage with the lab at a pace that is appropriate for their abilities and interests.

Science: Grade 7

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Term 2

WEEK 2	⬚ **30m Natural History: Grade 7 - Lesson 14** *Molting* ▯ Materials: The Girl Who Drew Butterflies **→ RECAP** What did you read or observe last week? **→ READ, NARRATE, & DISCUSS** Ch.7 p.41-44 "Although tulip" - "to a branch." **→ READ, NARRATE, & DISCUSS** p.46-47 "As Maria discovered" - "fully understood."	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 2	⬚ **30m General Science: Grade 7 - Lesson 27** *Aristotle* ▯ Materials: Aristotle Leads the Way **→ RECAP** What did you read last time? **→ READ, NARRATE, & DISCUSS** Ch.11 p.100-105 "The discussed" - "good thinkers do." • Compare/contrast Plato and Aristotle. • How does this chapter connect back to ideas from previous chapters? At the same time, how does it move the story forward?	
WEEK 2	⬚ **30m General Science: Grade 7 - Lesson 28** *Aristotle's Incorrect-Yet-Important Model of Earth* ▯ Materials: Aristotle Leads the Way **→ RECAP** What did you read last time? **→ READ, NARRATE, & DISCUSS** Ch.12 p.106-111 "Funny thing" - "what they had." • Notice again the theme that even incorrect ideas can be important. Discuss. • Was Aristotle a genius who moved science forward, or was he responsible for holding science back?	
WEEK 2	⬚ **45m Labs: Grade 7 - Lesson 14** *Engineering a Catapult* ▯ Materials: Grade 7 Lab Book and materials listed within **→ LAB DAY** Complete day 2 of Engineering a Catapult, as directed in the Lab Book. Suggested day 2: Students complete the Procedure today.	
WEEK 3	⬚ **30m Natural History: Grade 7 - Lesson 15** *Judge Not Too Quickly* ▯ Materials: The Girl Who Drew Butterflies **→ RECAP** What did you read or observe last week?	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Science: Grade 7

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Term 2

	→ READ, NARRATE, & DISCUSS Ch.7 p.48-54 "Sometimes" - "lot of effort."	
WEEK 3	**30m General Science: Grade 7 - Lesson 29** *Wandering in the Sky* Materials: Aristotle Leads the Way **→ RECAP** What did you read last time? **→ READ, NARRATE, & DISCUSS** Ch.12 p.112-113 "If you happen" - "merry-go-round." • Explain why planetary movement is confusing to a sky watcher on Earth.	**• NATURE NOTEBOOK** Prompt for General Science in Outdoor Work Quick Link.
WEEK 3	**30m General Science: Grade 7 - Lesson 30** *A Very Different Model of Earth* Materials: Aristotle Leads the Way **→ RECAP** What did you read last time? **→ READ, NARRATE, & DISCUSS** Ch.13 p.114-117 "Aristotle's universe" - "their big ideas." • What's the 'big idea' or 'main purpose' of this chapter? • How does this chapter connect back to ideas from previous chapters? At the same time, how does it move the story forward?	**• IMPORTANT DATES** Aristarchus (c.310 B.C.-c.230 B.C.)
WEEK 3	**45m Labs: Grade 7 - Lesson 15** *Engineering a Catapult* Materials: Grade 7 Lab Book and materials listed within PREP: Read Teacher Tip **→ LAB DAY** Complete day 3 of Engineering a Catapult, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook. Suggested day 3: Students complete the Analysis and Conclusion.	★ TEACHER TIP Be sure to have students share their notebooks, during which they should be able to discuss the science and explain what they did.
WEEK 4	**30m Natural History: Grade 7 - Lesson 16** *Pupa* Materials: The Girl Who Drew Butterflies **→ RECAP** What did you read or observe last week? **→ READ, NARRATE, & DISCUSS** Ch.8 p.55-60 "Encouraged" - "Waltha Castle." • In this reading, we heard that Maria did not have a good relationship with her husband. Think about some healthy relationships that you have seen in your family or friends. Talk with a trusted adult or write about what makes a relationship healthy.	★ TEACHER NOTE Maria Merian's estrangement from her husband is discussed on p.58-59. Teachers might choose to allow it to pass by the way or discuss it with students. **• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Science: Grade 7

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Term 2

WEEK 4	⬚ **30m General Science: Grade 7 - Lesson 31** *How Far the Moon?* Materials: Aristotle Leads the Way **➔ RECAP** What did you read last time? **➔ READ, NARRATE, & DISCUSS** Ch.13 p.118-119 "How do you" - "came around." ● How do you use time to measure distance (without a laser)? **➔ SUPPLEMENTAL** ∞ Video Link: How Levers Work (4:46)	**● NATURE NOTEBOOK** Prompt for General Science in Outdoor Work Quick Link.
WEEK 4	⬚ **30m General Science: Grade 7 - Lesson 32** *The City of Alexandria* Materials: Aristotle Leads the Way **➔ RECAP** What did you read last time? **➔ READ, NARRATE, & DISCUSS** Ch.14 p.120-125 "The Mediterranean" - "astonishing city." ● Tell about Alexandria. Consider how Alexandria connects back to previous chapters and why it might be important moving forward.	**● IMPORTANT DATES** Ptolemaic Dynasty (332 B.C-30 A.D.), City of Alexandria (est. 331 B.C.)
WEEK 4	⬚ **45m Labs: Grade 7 - Lesson 16** *Engineering a Complex Machine* Materials: Grade 7 Lab Book and materials listed within PREP: Read Teacher Tip **➔ LAB DAY** Complete day 1 of Engineering a Complex Machine, as directed in the Lab Book. Suggested day 1: Students complete the Introduction, gather materials, and begin step 1 of the Procedure.	★ TEACHER TIP Students construct a device with several simple machines combined.
WEEK 5	⬚ **30m Natural History: Grade 7 - Lesson 17** *Eclosing* Materials: The Girl Who Drew Butterflies PREP: Read Teacher Tip **➔ RECAP** What did you read or observe last week? **➔ READ & NARRATE** Ch.9 p.61-64, 66-67 "Windswept" - "scientific legacy." **➔ READ, NARRATE, & DISCUSS** p.65 "In the 1600s" - "marriage." ● How should faith support our relationship with God, other people, and the world around us?	★ TEACHER NOTE Some aspects of 17th-century religious practices/beliefs are mentioned on p.65, including the ordering of society, evolution, and marriage. Teachers should pre-read if they have concerns. ★ TEACHER TIP Are your learners demonstrating more familiarity/interest in plants and ecosystem relationships? More interest/ease in discussing the relationships that plants have with the rest of the ecosystem? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a nature preserve or natural history museum.

Science: Grade 7

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Term 2

● **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5 ⬚ **30m General Science: Grade 7 - Lesson 33**
The Pharos Lighthouse

Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

➔ **RECAP**
What did you read last time?

➔ **READ, NARRATE, & DISCUSS**
Ch.14 p.126-127 "A gigantic" - "Mediterranean Sea."
● Tell about the Pharos lighthouse.

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in engineering or how Things work? More interest/ease in discussing the complex relationship between science and society? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a play ground or science museum with an engineering exhibit. Reach out through Contact Us, if you need help.

● **NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

WEEK 5 ⬚ **30m General Science: Grade 7 - Lesson 34**
Technology in Alexandria

Materials: Aristotle Leads the Way

➔ **RECAP**
What did you read last time?

➔ **READ, NARRATE, & DISCUSS**
Ch.15 p.128-133 "For centuries" - "to jest with."
● Describe technology at this time.
● Tell about Hero.

➔ **SUPPLEMENTAL**
∞ Video Link: Human Powered Helicopter (5:36)

● **FIELD TRIP**
Visit a playground and notice the simple machines with which the equipment is designed.

● **IMPORTANT DATES**
Hero (1st century A.D.)

WEEK 5 ⬚ **45m Labs: Grade 7 - Lesson 17**
Engineering a Complex Machine

Materials: Grade 7 Lab Book and materials listed within

➔ **LAB DAY**
Complete day 2 of Engineering a Complex Machine, as directed in the Lab Book.

Suggested day 2: Students conduct most of the Procedure today. Depending on time, they may choose to leave step 3 until next week.

WEEK 6 ⬚ **30m Natural History: Grade 7 - Lesson 18**
Trouble

Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

➔ **RECAP**
What did you read or observe last week?

★ TEACHER TIP
The next 2 weeks are activity weeks!

★ TEACHER NOTE
The conclusion of Merian's unhappy marriage is presented on p.72: "Then trouble came" - "again a Merian." Teachers might

Term 2

→ **READ, NARRATE, & DISCUSS**
Ch.9 p.67, 70-73 "Spring came" - "to Amsterdam."

→ **READ, NARRATE, & DISCUSS**
p.68-69 "Maria called" - "for camouflage."

choose to allow it to pass by the way or to engage in discussion.

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6 ⬚ **30m General Science: Grade 7 - Lesson 35**
Hero's Air-Powered Machines

Materials: Aristotle Leads the Way

→ **RECAP**
What did you read last time?

→ **READ, NARRATE, & DISCUSS**
Ch.15 p.134-135 "The title" - "self-trimming wick."
• What's a pneumatic device?

• **NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

WEEK 6 ⬚ **30m General Science: Grade 7 - Lesson 36**
Euclid, Philosopher of Math

Materials: Aristotle Leads the Way

→ **RECAP**
What did you read last time?

→ **READ, NARRATE, & DISCUSS**
Ch.16 p.136-142 "A math professor" - "what I mean."
• Explain how Euclid took math "further" than business dealings and measurement.

• **IMPORTANT DATES**
Euclid (c.325 B.C.-c.270 B.C.)

WEEK 6 ⬚ **45m Labs: Grade 7 - Lesson 18**
Engineering a Complex Machine

Materials: Grade 7 Lab Book and materials listed within

PREP: Read Teacher Tip

→ **LAB DAY**
Complete day 3 of Engineering a Complex Machine, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 3: Students complete the lab this week.

★ TEACHER TIP
Be sure to have students share their notebooks, during which they should be able to discuss the science and explain what they did.

WEEK 7 ⬚ **30m Natural History: Grade 7 - Lesson 19**
Botanical Art Activity

Materials: The Girl Who Drew Butterflies, paper, paint, field guides, nature journal, or other records of observation

PREP: Read Teacher Tip

→ **INTRO**
Over the next 2 weeks, create botanical art in the style of Maria Merian using what you have learned about plant anatomy and its relationship with an insect. Use whatever medium best suits you. You can even use a computer!

→ **ACTIVITY**

★ TEACHER TIP
Interested students may need additional time during their composition period this week and/or next to complete this in-class assignment.

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Science: Grade 7

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Term 2

WEEK 7	**30m General Science: Grade 7 - Lesson 37** *Thinking about Prime Numbers* Materials: Aristotle Leads the Way PREP: Read Teacher Tip **→ RECAP** What did you read last time? **→ READ, NARRATE, & DISCUSS** Ch.16 p.143-145 "Do you like" - "of as 'molecules.'" • Explain prime numbers. • Try Eratosthenes' Sieve as described on page 145. How many primes can you find?	★ TEACHER TIP Remember to engage students on their reading through natural discussion. If you need help, use the bulleted prompts. **• NATURE NOTEBOOK** Prompt for General Science in Outdoor Work Quick Link.
WEEK 7	**30m General Science: Grade 7 - Lesson 38** *Engineering to Solve Problems* Materials: Aristotle Leads the Way **→ RECAP** What did you read last time? **→ READ, NARRATE, & DISCUSS** Ch.17 p.146-152 "It sounded" - "I have it!" • If Hero was an engineer and Euclid was a mathematician, what was Archimedes? • Archimedes seemed to have some mixed feelings about engineering. Explain. How do you think the ancients would feel about the trend toward biomimicry in our modern era? **→ SUPPLEMENTAL** Article Resource: Fascinating World of Biomimicry (5:59)	★ TEACHER NOTE The age of the Earth is mentioned in the supplemental video (2:50 and 5:12). Teachers might choose to allow it to pass by the way, to discuss with students, or to skip this one. **• IMPORTANT DATES** Archimedes (c.287 B.C.-212 B.C.)
WEEK 7	**45m Labs: Grade 7 - Lesson 19** *Engineering Challenge* Materials: Grade 7 Lab Book and materials listed within PREP: Read Teacher Tip **→ LAB DAY** Complete day 1 of the Engineering Challenge, as directed in the Lab Book. Suggested day 1: Students complete the Introduction and gather materials today.	★ TEACHER TIP Students design, construct, and test their own complex machine. Students should be able to find everything they need from household and recycled items, but teachers may want to check in to see if any materials are needed for next week.
WEEK 8	**30m Natural History: Grade 7 - Lesson 20** *Botanical Art Activity* Materials: The Girl Who Drew Butterflies, paper, paint, field guides, nature journal, or other records of observation **→ ACTIVITY** Complete your botanical art.	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Science: Grade 7

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Term 2

WEEK 8 ⬚ **30m General Science: Grade 7 - Lesson 39**
Engineering to Solve Problems

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.17 p.152-157 "When Archimedes" - "of his brain."
- Tell the story of "Eureka!"
- Archimedes solved problems by looking at them differently. Discuss.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 8 ⬚ **30m General Science: Grade 7 - Lesson 40**
How did the Claw Really Work?

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.17 p.158-159 "Did Archimedes" - "to find it."
- Why is the claw questioned when there are historical accounts of it?

WEEK 8 ⬚ **45m Labs: Grade 7 - Lesson 20**
Engineering Challenge

Materials: Grade 7 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of the Engineering Challenge, as directed in the Lab Book.

Suggested day 2: Students work on their design this week.

WEEK 9 ⬚ **30m Natural History: Grade 7 - Lesson 21**
Expanding

Materials: The Girl Who Drew Butterflies

→ RECAP
What did you read or observe last week?

→ READ, NARRATE, & DISCUSS
Ch.10 p.74-77 "After Maria's" - "or two."

→ READ, NARRATE, & DISCUSS
p.78-79 "Ever since" - "about evolution."

★ TEACHER NOTE
Evolution is mentioned toward the end of p.78. Teachers might choose to allow it to pass by the way or decide how they would like to discuss it.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9 ⬚ **30m General Science: Grade 7 - Lesson 41**
Measuring the Earth

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.18 p.160-163 "Eratosthenes" - "all about it."
- What's the 'big idea' or 'main purpose' of this chapter?
- How does this chapter connect back to ideas from previous chapters?

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

• IMPORTANT DATES
Eratosthenes (c.275 B.C.-c.195 B.C.)

Term 2

At the same time, how does it add to the story?

➔ SUPPLEMENTAL
∞ Video Link: Six Simple Machines in One (7:08)

WEEK 9 **30m General Science: Grade 7 - Lesson 42**
How did Eratosthenes Do It?

Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.18 p.164-165 "Syne, near modern" - "24,855 miles."
● Explain how Eratosthenes found the circumference of the Earth.

WEEK 9 **45m Labs: Grade 7 - Lesson 21**
Engineering Challenge

Materials: Grade 7 Lab Book and materials listed within

➔ LAB DAY
Complete day 3 of the Engineering Challenge, as directed in the Lab Book.

Suggested day 3: Students finish and test their design this week. If they want an extra week, there is a catch-up day during the last week.

WEEK 10 **30m Natural History: Grade 7 - Lesson 22**
Preparing

Materials: The Girl Who Drew Butterflies

➔ RECAP
What did you read or observe last week?

➔ READ, NARRATE, & DISCUSS
Ch.10 p.80-83 "Maria was" - "unknown land?"

● OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10 **30m General Science: Grade 7 - Lesson 43**
The Roman Empire

Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.19 p.166-169 "It was the" - "to be controlled."
● Discuss some positive and negative aspects of the shift from Greek to Roman prominence.
● How might the work of the Greeks from the last few chapters have been different in Roman culture?

● NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

● IMPORTANT DATES
Library of Alexandria burned (48 B.C.)

WEEK 10 **30m General Science: Grade 7 - Lesson 44**
The Roman Empire

Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

● IMPORTANT DATES
Strabo (c.64 B.C.-c.24 A.D.)

Science: Grade 7

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Term 2

→ READ, NARRATE, & DISCUSS
Ch.19 p.169-173 "Jerusalem was" - "Brueghel the Elder."
• Tell about Strabo.

WEEK 10 **45m Labs: Grade 7 - Lesson 22**
Engineering Challenge

Materials: Grade 7 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 4 of the Engineering Challenge, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 4: Students complete the lab this week.

★ TEACHER TIP
Be sure to have students share their notebooks, during which they should be able to discuss the science and explain what they did.

WEEK 11 **30m Natural History: Grade 7 - Lesson 23**
Catch Up

Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

→ CATCH UP
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

★ TEACHER TIP
When students take exams next week, evaluate what they recall, but more importantly, whether plants or the relationships that plants have with the rest of the ecosystem are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11 **30m General Science: Grade 7 - Lesson 45**
Mapping Sky and Land

Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.20 p.174-180 "It's easy" - "called triangulation."
• What's the 'big idea' or 'main purpose' of this chapter?
• How do the ideas connect back to previous chapters? At the same time, how does it move the story forward? What does the science seem to be building toward?

→ SUPPLEMENTAL
∞ Video Link: Precession of Earth (3:13)

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly whether engineering, design, or the relationship between science and society are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? Reach out through Contact Us, if you need help.

• IMPORTANT DATES
Hipparchus (c.190 B.C.-c.120 B.C.)

Science: Grade 7

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Term 2

WEEK 11	⬚ **30m General Science: Grade 7 - Lesson 46** *Points of Reference* ▯ Materials: Aristotle Leads the Way **➔ RECAP** What did you read last time? **➔ READ, NARRATE, & DISCUSS** Ch.20 p.181-183 "A point" - "(see page 207)." • Describe how to chart stars.	**• NATURE NOTEBOOK** Prompt for General Science in Quick Link.
WEEK 11	⬚ **45m Labs: Grade 7 - Lesson 23** *Catch Up* ▯ Materials: Grade 7 Lab Book and materials listed within **➔ CATCH UP** Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	
WEEK 12	⬚ **30m Natural History: Grade 7 - Lesson 24** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: The Girl Who Drew Butterflies	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 12	⬚ **30m General Science: Grade 7 - Lesson 47** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: Aristotle Leads the Way	
WEEK 12	⬚ **30m General Science: Grade 7 - Lesson 48** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: Aristotle Leads the Way	
WEEK 12	⬚ **45m Labs: Grade 7 - Lesson 24** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: Grade 7 Lab Book	

Science: Grade 7

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Term 3

WEEK 1

⬚ **30m Natural History: Grade 7 - Lesson 25**

Flight

▯ Materials: The Girl Who Drew Butterflies

→ RECAP
What did you read or observe last week?

→ READ, NARRATE, & DISCUSS
Ch.11 p.84-87 "In late" - "breeding caterpillars."

→ READ, NARRATE, & DISCUSS
p.88-89 "The history" - "in 1863."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1

⬚ **30m General Science: Grade 7 - Lesson 49**

Ptolemy the Thinker

▯ Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ NOTE
The last three lessons of the year have been left open to complete unfinished readings, labs, projects, or science notebooks.

→ RECAP
Recall where we left the story. Look back if you need help.

→ READ, NARRATE, & DISCUSS
Ch.21 p.184-189 "Claudius Ptolemy" - "out of style."
• A lot of Ptolemy's ideas were incorrect. Why is he remembered as a great thinker?

→ SUPPLEMENTAL
∞ Video Link: Armillary Sphere (3:42)
∞ Video Link: Animation of How the Sphere Works (4:14)

★ TEACHER TIP
Remember to engage students on their reading through natural discussion. If you need help, use the bulleted prompts.

• IMPORTANT DATES
Claudius Ptolemy (c.85 A.D.-c. 165 A.D.)

WEEK 1

⬚ **30m General Science: Grade 7 - Lesson 50**

The Material Versus the Spiritual

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.22 p.190-193 "After some" - "create new cultures."
• Describe the cultural change that occurred as society shifted from Roman to Christian philosophy.
• Consider the quote, "The very struggle to become a Christian... seemed to be a struggle against astrology." How might the reaction to science by some Church leaders have been influenced by their reaction to the materialistic thinking of the Greeks and Romans?

WEEK 1

⬚ **45m Labs: Grade 7 - Lesson 25**

The Circumference of the Earth

▯ Materials: Grade 7 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

★ TEACHER TIP
Students evaluate the circumference of the Earth empirically. To do so, they must coordinate with a person in a different location and the weather. There is flexibility to push back a week, if needed.

Term 3

Complete day 1 of The Circumference of the Earth, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction. Be sure that students gather materials, complete step 1 of the Procedure, and coordinate with their partner before the day of data collection. Data collection should be on a sunny day, so they may need some flexibility to adjust for weather, if necessary.

Pacing is only a suggestion. Students should engage with the lab at a pace that is appropriate for their abilities and interests.

WEEK 2

30m Natural History: Grade 7 - Lesson 26

New Lands

Materials: The Girl Who Drew Butterflies, video links

→ RECAP
What did you read or observe last week?

→ READ, NARRATE, & DISCUSS
Ch.11 p.90-95 "Maria quickly" - "of sugarcane."

→ SUPPLEMENTAL
Scientists who study butterflies have been inspired in many ways:
∞ Video Link: Biomimicry and Butterflies
∞ Video Link: Butterfly-Inspired Robots

★ TEACHER NOTE
The experience of native and enslaved women, including suicide and the use of a plant to abort pregnancy, is mentioned on p.90-91: "The Indians" - "customs and lore." Teachers might choose to allow it to pass by the way, contributing to students' knowledge from history lessons, or engage students in discussion.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2

30m General Science: Grade 7 - Lesson 51

The Material Versus the Spiritual

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.22 p.193-197 "In the meantime," - "barbarians had won."
• Contrast Cyril's and Augustine's reactions to the events and thinking of their time. Why do you think they reacted so differently?

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

• IMPORTANT DATES
Augustine (354 A.D.-430 A.D.)

WEEK 2

30m General Science: Grade 7 - Lesson 52

Thinking about Integrity

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.22 p.198-199 "Caius Julius Solinus" - "and Essence"
• What was the Polyhistor?

WEEK 2

45m Labs: Grade 7 - Lesson 26

The Circumference of the Earth

Materials: Grade 7 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of The Circumference of the Earth, as directed in the Lab Book.

Suggested day 2: Students complete the Procedure today. This should be a sunny day, and their partner should be collecting data, as well! If they want an extra week for this or any other lab, there is a catch-up day during the last week.

WEEK 3 ⬚ **30m Natural History: Grade 7 - Lesson 27**

New Ideas

▯ Materials: The Girl Who Drew Butterflies

→ RECAP
What did you read or observe last week?

→ READ, NARRATE, & DISCUSS
Ch.11 p.96-102 "Life was" - "had never seen."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3 ⬚ **30m General Science: Grade 7 - Lesson 53**

Rejection of the Greeks

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.23 p.200-205 "In 1529," - "improved on Ptolemy."
• What do you think about the term "Dark Ages"? Is it an accurate representation of the period or not?
• Some writers have suggested that society needed a period of penitence after Greek and Roman rule. Discuss this idea. Why would this be needed? What would such penitence look like?
• Charlotte Mason advised (Vol 3 p.49), 'we must bear in mind that truth behaves like a country gate allowed to 'swing to' after a push. Now it swings a long way to this side and now a long way to that, and at last, after shorter and shorter oscillations, the latch settles.' If she is right, what do you expect will happen next in the Story of Science?

• IMPORTANT DATES
Middle Ages (476 A.D.-1000 A.D.)

WEEK 3 ⬚ **30m General Science: Grade 7 - Lesson 54**

The Gate Swings

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.24 p.206-209 "Pope Sylvester II" - "not encouraged."
• What's the 'big idea' or 'main purpose' of this chapter?
• How do the ideas connect back to previous chapters? At the same time, how does it add to the story?

→ SUPPLEMENTAL
∞ Video Link: How to Use an Astrolabe (4:56)
∞ Video Link: Beauty of an Astrolabe (9:09)

WEEK 3 ⬚ **45m Labs: Grade 7 - Lesson 27**

The Circumference of the Earth

▯ Materials: Grade 7 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

★ TEACHER TIP
Be sure to have students share their notebooks, during which they should be able to discuss the science and explain what they did.

Term 3

Complete day 3 of The Circumference of the Earth, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 3: Students complete the Analysis and Conclusions.

WEEK 4

⬚ **30m Natural History: Grade 7 - Lesson 28**

Egg

Materials: The Girl Who Drew Butterflies

→ RECAP
What did you read or observe last week?

→ READ, NARRATE, & DISCUSS
Ch.12 p.103-109 "By the time" - "on details."
● This chapter is titled "Egg," just like the first. What do you think is the egg that Maria left?

● OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 4

⬚ **30m General Science: Grade 7 - Lesson 55**

The Gate Swings

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.24 p.210-217 "Around 1110," - "called samurai."
● Discuss the role of learning and education in society. Is learning a basic human need? What role does curiosity hold? Should all ideas be completely free and available for everyone?

● IMPORTANT DATES
Adelard translates Euclid into Latin (c.1100 A.D.)

WEEK 4

⬚ **30m General Science: Grade 7 - Lesson 56**

Mathematical Revolution

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.25 p.218-221 "Do you want" - "handle its basics."
● Explain: "The Indian thinkers took three ideas that had been used elsewhere and made a powerful connection between them."

→ SUPPLEMENTAL
∞ Video Link: Why Can't You Divide by Zero? (4:51)

● IMPORTANT DATES
Brahmagupta (c.598 A.D.-c.660 A.D.)

WEEK 4

⬚ **45m Labs: Grade 7 - Lesson 28**

Scale Solar System Model

Materials: Grade 7 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 1 of the Scale Solar System Model, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction and begin the Procedure today.

★ TEACHER TIP
Students design and construct a model of the solar system to scale. Remember, the most important objective is always building skills and habits. Pacing is only a suggestion.

Science: Grade 7

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Term 3

WEEK 5 ⬚ **30m Natural History: Grade 7 - Lesson 29**
Patience

Materials: The Girl Who Drew Butterflies

→ RECAP
What did you read or observe last week?

→ READ, NARRATE, & DISCUSS
Ch.12 p.110-115 "Published in" - "miraculous lives."

● Discuss what 'ecology' is, based on Maria Merian's work.

You can hear or read more at the following links:
∞ Video Link: Introduction to Ecology
∞ Video Link: Why do we need insects?

★ TEACHER NOTE
The age of the Earth is mentioned in the second supplemental video (0:13). Teachers might choose to allow it to pass by the way, to discuss, or to skip this one.

● **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5 ⬚ **30m General Science: Grade 7 - Lesson 57**
Mathematical Revolution

Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.25 p.221-225 "Someone had to" - "you do today."
● What's the 'big idea' or 'main purpose' of this chapter?
● How do the ideas connect back to previous chapters? At the same time, how does it move the story forward?

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in math or its relationship with science? More interest/ease in discussing the complex relationship between science and society? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a science museum. Reach out through Contact Us, if you need help.

● **IMPORTANT DATES**
House of Wisdom (est. 830 A.D.), Islamic Golden Age (800 A.D.-1258 A.D.), Leonardo Pisano [Fibonacci] (c.1170 A.D.-c.1250 A.D.)

WEEK 5 ⬚ **30m General Science: Grade 7 - Lesson 58**
Fibonacci's Numbers

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.25 p.226-227 "Fibonacci" - "not always exact."
● What are Fibonacci's numbers conceptually?

● **NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

WEEK 5 ⬚ **45m Labs: Grade 7 - Lesson 29**
Scale Solar System Model

Materials: Grade 7 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of the Scale Solar System Model, as directed in the Lab Book.

Suggested day 2: Students complete at least through step 4 of the Procedure today.

Science: Grade 7

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Term 3

WEEK 6

⬚ **30m Natural History: Grade 7 - Lesson 30**
In Her World

Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

→ RECAP
What did you read or observe last week?

→ READ, NARRATE, & DISCUSS
p.116-121 "Maria Sibylla" - "it forever."

• "She saw nature as an ever-transforming web of connections…" What types of connections have you noticed in nature? Perhaps there will be some new ones in the video link:
∞ Video Link: Feedback Loops and Networks in Nature

★ TEACHER TIP
Are your learners demonstrating more familiarity/interest in ecological relationships? More interest/ease in discussing the life and work of Maria Merian? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a nature preserve or natural history museum.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

• IMPORTANT DATES
Maria Merian (1647-1717)

WEEK 6

⬚ **30m General Science: Grade 7 - Lesson 59**
Aquinas and the Truce

Materials: Aristotle Leads the Way

PREP: Read Teacher Tip

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.26 p.228-232 "In 1225," - "job of it."
• Tell about Aquinas' life.
• Contrast learning in a monastery with learning in a university, according to the author. What scientific question is in the air at the university at this time?

★ TEACHER TIP
Remember to engage students on their reading through natural discussion. If you need help, use the bulleted prompts.

• IMPORTANT DATES
Thomas Aquinas (1225-1274 A.D.), Roger Bacon (c.1214-1292 A.D.)

WEEK 6

⬚ **30m General Science: Grade 7 - Lesson 60**
Aquinas and the Truce

Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.26 p.233-235 "In Paris," - "above it all."
• We have now seen men of faith both promote scientific thinking and caution against scientific thinking. Is faith at the root of this conflict? If so, explain. If not, what do you think is?

→ SUPPLEMENTAL
The Triumph of Saint Thomas Aquinas was the subject of more than one artist (the author mentions one version on p.235). Consider two of them at the links. What do you notice, and what does this have to do with his message about truth and learning?
∞ Image Link: Triumph by Gozzoli
∞ Image Link: Triumph by da Firenze

Science: Grade 7

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Term 3

WEEK 6

⬚ **45m Labs: Grade 7 - Lesson 30**
Scale Solar System Model

▯ Materials: Grade 7 Lab Book and materials listed within

➔ LAB DAY
Complete day 3 of the Scale Solar System Model, as directed in the Lab Book.

Suggested day 3: Students complete step 5 of the Procedure today.

WEEK 7

⬚ **30m Natural History: Grade 7 - Lesson 31**
An Ecological Story

▯ Materials: video link

PREP: Read Teacher Tip

➔ INTRO
Humans have an impact on the rest of the natural world. Because everything is ecologically connected, the full impact is often unexpected. Watch a short video during today's lesson.

➔ VIEW & DISCUSS
∞ Video Link: The Hopeful Story of a Doomed Fox
• What do you see as the pros and cons of human intervention in this story? What is our responsibility when we disrupt an ecosystem?

★ TEACHER TIP
Next week is an activity week (as is the rest of the term). Students will need supplies, including a freshwater weed. This can be a bit of water weed from a pond (see this week's outdoor work) or any plant sold for freshwater fish tanks.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7

⬚ **30m General Science: Grade 7 - Lesson 61**
Imagination Predicts Innovation

▯ Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.26 p.236-237 "Like many" - "glittering jewels."
• Describe some of the future inventions predicted by Roger Bacon. What kind of thinking allows scientists (or science fiction writers) to make such predictions? Are they predictions or are they self-fulfilling?

WEEK 7

⬚ **30m General Science: Grade 7 - Lesson 62**
The Power of Books

▯ Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.27 p.238-242 "Johannes Gutenberg" - "work of a genius."
• What's the 'big idea' or 'main purpose' of this chapter?
• How do the ideas connect back to previous chapters? What do you think will happen next?

• IMPORTANT DATES
Gutenberg prints first book on printing press (1445 A.D.)

Science: Grade 7

Click THIS text or scan the QR code for links.

Term 3

WEEK 7

⬚ **45m Labs: Grade 7 - Lesson 31**
Scale Solar System Model

▯ Materials: Grade 7 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 4 of the Scale Solar System Model, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 4: Students complete step 6 of the Procedure, as well as the Analysis and Conclusions.

★ TEACHER TIP
Be sure to have students share their notebooks, during which they should be able to discuss the science and explain what they did.

WEEK 8

⬚ **30m Natural History: Grade 7 - Lesson 32**
Photosynthesis Activity

▯ Materials: any freshwater weed, pH indicator, 4 tubes, holder for tubes such as a mug beaker or measuring cup, water, straw, aluminum foil, sunny window, microscope, slides and coverslips, methylene blue, table salt

→ INTRO
We return our focus to plants now, but with the deeper understanding of ecology that Maria Merian has taught us. One of the most amazing characteristics of plants is their ability to use the carbon dioxide gas that animals breathe out and, using the energy of sunlight, turn it into sugar and oxygen gas. This is called photosynthesis. Eventually, animals eat the plants and return the carbon to the atmosphere, completing the carbon cycle. Today, we examine chloroplasts, where photosynthesis happens in the plant, and even get to "see" them at work.

→ ACTIVITY
∞ Activity Link: Photosynthesis Activity
∞ Video Link: Wet Mount Slide of Elodea
∞ Video Link: Wet Mount with Salt Water

→ VIEW & NARRATE
At a convenient time later in the day, observe each of the 4 tubes. Do you notice any signs that the plant is using the carbon dioxide gas you added to the water? Do you notice any signs that the plant is releasing oxygen? Under what conditions? Draw or diagram your 4 tubes in your science notebook. If you want, you may return the tubes to their setup and observe one more time tomorrow. Why do you think it might be different tomorrow? Is there anything else you would like to investigate?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8

⬚ **30m General Science: Grade 7 - Lesson 63**
The Power of Books

▯ Materials: Aristotle Leads the Way

→ RECAP
What did you read last time?

→ READ, NARRATE, & DISCUSS
Ch.27 p.242-247 "We don't know" - "we be now?"
• How can people learn to differentiate between right and wrong, correct and incorrect ideas? Is it always bad to believe an idea even if it turns out to be incorrect? When is it okay and when is it not okay? How do we know?

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

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Term 3

	• What was the underlying source of tension between some scientific and religious leaders during the time period of this book? (Look back at Ch.5, 26, 27, if you need examples.)	
WEEK 8	⬚ **30m General Science: Grade 7 - Lesson 64** *What Magellan's Journey Meant* Materials: Aristotle Leads the Way **→ RECAP** What did you read last time? **→ READ, NARRATE, & DISCUSS** Ch.28 p.248-252 "On September 4," - "about the Americas." • What role does Magellan's accomplishment have in the story of science? • Was Magellan a scientist? Why or why not?	
WEEK 8	⬚ **45m Labs: Grade 7 - Lesson 32** *What Does All This Mean to Us?* Materials: Grade 7 Lab Book and materials listed within PREP: Read Teacher Tip **→ LAB DAY** Complete day 1 of What Does All This Mean to Us?, as directed in the Lab Book. Suggested day 1: Students complete the Introduction and gather materials today.	★ TEACHER TIP Students evaluate how physical features on Earth contribute to climate. They are invited to ponder Earth's design and whether this has any relevance to belief systems. Teachers who want to engage students in discussion should pre-read the Introduction.
WEEK 9	⬚ **30m Natural History: Grade 7 - Lesson 33** *Composition Activity* Materials: by student design **→ INTRO** Over the next 2 weeks, create a composition of any type you like that describes the importance of plants for other creatures. You might refer to what you learned from Maria Merian or other books, or what you have learned in any of our activities this year. You might compose a poem, a piece of art, an essay, a slide presentation, etc. **→ ACTIVITY**	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 9	⬚ **30m General Science: Grade 7 - Lesson 65** *What Magellan's Journey Meant* Materials: Aristotle Leads the Way **→ RECAP** What did you read last time? **→ READ, NARRATE, & DISCUSS** Ch.28 p.253-257 "A seasoned soldier" - "changed everything." • "Magellan changed everything." Explain.	**• IMPORTANT DATES** Magellan sails around the world (1519-1522 A.D.)

Science: Grade 7

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Term 3

WEEK 9 ⬚ **30m General Science: Grade 7 - Lesson 66**
Imagining a Journey

Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.29 p.258-265 "Far, far from" - "we were coming."
• What's the 'big idea' or 'main purpose' of this chapter?
• How do the ideas connect back to previous chapters? At the same time, how does it lead us forward?

★ TEACHER NOTE
A short science fiction story is presented in which the timescale of Earth's existence is important on p.261: "Thuleans send" - "of particular interest." Teachers might choose to allow it to pass by the way or to discuss with students.

WEEK 9 ⬚ **45m Labs: Grade 7 - Lesson 33**
What Does All This Mean to Us?

Materials: Grade 7 Lab Book and materials listed within

➔ LAB DAY
Complete day 2 of What Does All This Mean to Us?, as directed in the Lab Book.

Suggested day 2: Students complete the Procedure today.

WEEK 10 ⬚ **30m Natural History: Grade 7 - Lesson 34**
Composition Activity

Materials: by student design

➔ ACTIVITY
Complete your composition activity from last week.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10 ⬚ **30m General Science: Grade 7 - Lesson 67**
It's About Uncertainty

Materials: Aristotle Leads the Way

➔ RECAP
What did you read last time?

➔ READ, NARRATE, & DISCUSS
Ch.30 p.266-269 "There is something" - "the last word."
• "Science is not about certainty; it is about uncertainty." What do you think the author means by this? What does this mean for you?
• Why does the author claim that good scientists are humble? What other habits do you think they have?

★ TEACHER NOTE
The age of the Earth is mentioned on p.268: "Those two theories" - "fixed, permanent stage." Teachers might choose to allow it to pass by the way, to discuss with students, or to consider alternative theories they wish to share.

WEEK 10 ⬚ **30m General Science: Grade 7 - Lesson 68**
Catch Up

Materials: Aristotle Leads the Way

➔ CATCH UP
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

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Term 3

WEEK 10	**45m Labs: Grade 7 - Lesson 34** *What Does All This Mean to Us?* Materials: Grade 7 Lab Book and materials listed within PREP: Read Teacher Tip **➔ LAB DAY** Complete day 3 of What Does All This Mean to Us?, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook. Suggested day 3: Students complete the Analysis and Conclusions today.	★ TEACHER TIP Be sure to have students share their notebooks, during which they should be able to discuss the science and explain what they did.
WEEK 11	**30m Natural History: Grade 7 - Lesson 35** *Catch Up* Materials: The Girl Who Drew Butterflies PREP: Read Teacher Tip **➔ CATCH UP** Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	★ TEACHER TIP When students take exams next week, evaluate what they recall, but more importantly, whether ecological relationships or Maria Merian are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? **• FIELD TRIP** Visit a natural history museum butterfly exhibit or butterfly house.
WEEK 11	**30m General Science: Grade 7 - Lesson 69** *Catch Up* Materials: Aristotle Leads the Way **➔ CATCH UP** Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	**• NATURE NOTEBOOK** Prompt for General Science in Outdoor Work Quick Link.
WEEK 11	**30m General Science: Grade 7 - Lesson 70** *Catch Up* Materials: Aristotle Leads the Way PREP: Read Teacher Tip **➔ CATCH UP** Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	★ TEACHER TIP When student(s) take exams next week, evaluate what they recall, but more importantly whether math, its place in science, or the relationship between science and society are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? Reach out through Contact Us, if you need help.

Science: Grade 7

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Term 3

WEEK 11	⬚ **45m Labs: Grade 7 - Lesson 35** *Catch Up* Materials: Grade 7 Lab Book and materials listed within **➔ CATCH UP** Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	
WEEK 12	⬚ **30m Natural History: Grade 7 - Lesson 36** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: The Girl Who Drew Butterflies	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 12	⬚ **30m General Science: Grade 7 - Lesson 71** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: Aristotle Leads the Way	
WEEK 12	⬚ **30m General Science: Grade 7 - Lesson 72** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: Aristotle Leads the Way	
WEEK 12	⬚ **45m Labs: Grade 7 - Lesson 36** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: Grade 7 Lab Book	

Science: Grade 7

Examination

Term 1

General Science: Grade 7

· Aristotle Leads the Way: Explain at least two examples of how careful record-keeping and/or measurement advanced science in the ancient world.

· Aristotle Leads the Way: What were some obstacles that made science difficult for early scientists, and what urged them forward?

· Explain how you did one laboratory investigation this term and what you learned from it.

Natural History: Grade 7

· The Girl Who Drew Butterflies: Draw a diagram illustrating how plant anatomy is dependent on animal life and environmental conditions.

Term 2

General Science: Grade 7

· Aristotle Leads the Way: We are told that “Plutarch said that Archimedes believed engineering was ‘sordid and ignoble, [as is] every sort of art that lends itself to mere use and profit.’” Why then did he design so many useful devices? Describe 2 of them.

· Aristotle Leads the Way: Explain why Alexandria was different from other cities of its time. How did it help to create a new age of technological and intellectual advancement?

· Explain how you did one laboratory investigation this term and what you learned from it.

Natural History: Grade 7

· The Girl Who Drew Butterflies: Explain how close observation revealed new ideas to Maria that others did not see.

Term 3

General Science: Grade 7

· Aristotle Leads the Way: Explain how developments in mathematics changed the way scientists learned about their world.

· Aristotle Leads the Way: Were the Dark Ages really intellectually dark? Give reasons to support your conclusion.

· Explain how you did one laboratory investigation this term and what you learned from it.

Natural History: Grade 7

· The Girl Who Drew Butterflies: Explain how animals are ecologically dependent on plants. OR Give reasons why Maria Merian is sometimes called the first ecologist.